ECE 329 **Homework 3** **Due: Thursday, September 17, 2026, 11:59:00PM**

- Homeworks are due Thursday at 11:59:00 p.m. Late homework will not be accepted.
- Homeworks are to be turned in to Gradescope.
- Any deviation from the following steps will result in a 5 point penalty:
  - Write your name, netID, and section on first page of your submission.
  - Start each new problem on a new page.
  - Scan the homework as a PDF (rather than taking pictures). Use a free scanning app if you do not have access to a photocopier.
  - Upload the PDF to Gradescope and tag each problem's location in the PDF.
- Each student must submit individual solutions for each homework. You may discuss homework problems with other students registered in the course, but you may not copy their solutions. If you use any source outside of class materials that we've provided, you must cite every source that you used.
- Use of homework solutions from past semesters is not allowed and is considered cheating. Copying homework solutions from another student or generative AI is considered cheating.
- Penalties for cheating on homework: a zero for the assignment on the first offense, and an F in the course on the second offense.

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Reading Assignment: Kudeki: Lectures 6-9

1. Consider a simplified model of a **vacuum diode** consisting of a **cathode** in the $x = 0$ plane and an **anode** in the $x = d$ plane, where the anode is held to a constant potential $V_a = 2$ V relative to the cathode. The region $0 < x < d$ between the cathode and the anode supports a charge density $\rho(x)$ accounting for the electrons in transit from the cathode (where they are emitted) to the anode. If the potential distribution in the region $0 < x < d$ is given by $V(x) = V_a(\frac{x}{d})^{4/3}$, find the following:

   a) What is the velocity of an electron $v_e(\frac{x}{d})$ as a function of position in the range $0 < x < d$? Hint: remember what $V(x)$ means, physically.

   b) Plot the electric field $E_x(x)$,

   c) Plot the volumetric charge density $\rho(x)$,

   d) Determine the surface charge density ρ_s on the anode.

2. Given that $V(x, y, z) = x^2 - yz$ and $\mathbf{E} = -\nabla V$, what is $\nabla \times \mathbf{E}$?

3. An important vector identity which is true for any vector field $\mathbf{A}(x, y, z)$ is

$$\nabla \times (\nabla \times \mathbf{A}) = \nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A},$$

   where

$$\nabla^2 \mathbf{A} \equiv (\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2})\mathbf{A}$$

   is the *Laplacian* of $\mathbf{A}$ and $\nabla(\nabla \cdot \mathbf{A})$ is the gradient of the divergence of $\mathbf{A}$.

   Verify the identity for $\mathbf{A} = (x - y)\,\hat{x} + (x + y)\,\hat{y} + 2\,\hat{z}$ by calculating each side of the identity and showing them to be the same.

4. In a given region, $\mathbf{E} = 3x\,\hat{x} + y\,\hat{y} + 2z^2\,\hat{z}$ $\frac{\text{V}}{\text{m}}$.

   a) Determine the electrostatic potential $V(1, 2, 3)$ if $V(0, 0, 0) = 0$.

   b) What is the charge density ρ at the points $(0, 0, 0)$ and at $(1, 2, 3)$?

5. Given the fields $\mathbf{E_1} = y\hat{x} - x\hat{y}$ $\frac{\text{V}}{\text{m}}$ and $\mathbf{E_2} = y\hat{x} + x\hat{y}$ $\frac{\text{V}}{\text{m}}$, determine the circulation $\oint_C \mathbf{E} \cdot d\mathbf{l}$ for both $\mathbf{E_1}$ and $\mathbf{E_2}$ along a triangular path C traversing in order its vertices at $(x, y, z) = (-1, -1, 0)$, $(-1, 1, 0)$, and $(1, 1, 0)$ m.

   Hint: $d\mathbf{l} = (-\hat{x} - \hat{y})\,dx$ and $x = y$ on the slant edge of C.

6. Consider a static volumetric charge density $\rho(x, y, z) = 6\delta(y) + \rho_s\delta(y - 6)$ $\frac{\text{C}}{\text{m}^3}$ in a given region of free space (having permittivity ϵ_0), where the displacement field is $\mathbf{D} = \hat{x}\,2\epsilon_0 + \hat{y}\,4\epsilon_0$ $\frac{\text{C}}{\text{m}^2}$ for $0 < y < 6$ m and $D_y = 2\epsilon_0$ $\frac{\text{C}}{\text{m}^2}$ for $y > 6$ m. Furthermore, field $\mathbf{D}$ is uniform in each of regions $y < 0$, $0 < y < 6$ m, and $y > 6$ m.

   a) Determine ρ_s,

   b) Determine $\mathbf{D}$ for the region $y > 6$ m,

   c) Determine $\mathbf{D}$ for the region $y < 0$.

   d) Determine $\mathbf{E}$ in all three regions ($y < 0$, $0 < y < 6$, and $y > 6$)

   e) What is the voltage drop from the $y = 0$ m plane to the $y = 6$ m plane?

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f) Which of your answers above (parts a, b, c, d, or e) would change if the region from $0 < y < 6$ m were filled with a perfect dielectric having permittivity $\epsilon = 4\epsilon_0$ instead of a vacuum having ϵ_0? Explain.

7. A metal sphere of radius a carrying an electric charge Q is suspended in free space. It is surrounded, out to radius b, by linear dielectric material of permittivity ϵ and electric susceptibility χ.

   a) Determine $\mathbf{D}$ and $\mathbf{E}$ for the region $r < a$.

   b) Determine $\mathbf{D}$ and $\mathbf{E}$ for the region $a < r < b$. **Hint:** The free charge on the conducting sphere will distribute uniformly on its surface. This uniformity implies that the fields will be radial and spherically symmetric, $\mathbf{D} = D(r)\hat{r}$, $\mathbf{E} = E(r)\hat{r}$.

   c) Determine $\mathbf{D}$ and $\mathbf{E}$ for the region $r > b$.

   d) Determine the electric potential (relative to $r = \infty$) at the center of the sphere.

   e) Determine the polarization $\mathbf{P}$ and bound charge density ρ_b inside the dielectric.

   f) Determine the bound surface charge ρ_s on the inside and outside surface of the dielectric. Comment on how the bound surface charge on the inside edge of the dielectric affects the field.

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